SOVEREIGNAI — SYNTHETIC INDUSTRIAL INSPECTION REPORT

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Document ID: DEMO-INSP-2026-001 **Revision:** 1.0

Facility: Synthetic Process Unit A **Inspection Date:** 30 August 2026

Equipment: P-101 Process Pump **Inspection Type:** Routine Mechanical Inspection

Purpose: This synthetic report is designed to exercise document extraction, page-aware retrieval, inspection finding extraction, SOP retrieval, risk assessment, recommendation generation, citations, and approval-note generation.

1. Executive Summary

During a routine inspection of Process Pump P-101, the bearing temperature was recorded above the normal operating limit. Abnormal vibration was also observed. The bearing area showed signs of inadequate lubrication. The equipment was kept out of normal service pending inspection and corrective action.

2. Inspection Observations

Parameter	Observed	Applicable Normal Limit	Status
Bearing temperature	92 °C	Up to 80 °C	ABNORMAL
Vibration	6.8 mm/s RMS	Up to 4.5 mm/s RMS	ABNORMAL
Lubrication condition	Low / insufficient grease observed	Adequate lubrication required	ABNORMAL
Oil leakage	No visible leakage	No visible leakage	NORMAL

3. Detailed Findings

Finding F-001 — Bearing temperature exceeded the normal operating limit.

Observed bearing temperature was 92 °C. The applicable synthetic maintenance SOP states that normal bearing operating temperature is up to 80 °C. The reading therefore exceeds the documented limit by 12 °C.

Finding F-002 — Abnormal vibration was recorded.

Vibration at the pump bearing housing was measured at 6.8 mm/s RMS, compared with the synthetic inspection criterion of 4.5 mm/s RMS. The inspection team recorded the observation for follow-up.

Finding F-003 — Lubrication condition was inadequate.

Visual inspection indicated low/insufficient grease at the bearing lubrication point. The condition was recorded together with the equipment identifier and inspection date.

4. Immediate Actions Taken

• The observed temperature and vibration values were recorded against equipment P-101.

• P-101 was not returned to normal service while the abnormal bearing condition was under review.

• The affected maintenance area was identified for controlled inspection.

• Maintenance personnel were instructed to follow the applicable isolation and lockout/tagout procedure before intrusive maintenance.

5. Inspection Method

The inspection included review of equipment condition, bearing temperature measurement, vibration measurement, lubrication-condition observation, and visual inspection for leakage and other operational anomalies.

6. Synthetic Evidence for Knowledge-Base Matching

This section intentionally uses terminology aligned with the companion synthetic SOP documents so that a RAG/agent demonstration can retrieve authoritative evidence. The report does not contain the SOP itself; the agent is expected to retrieve the separate knowledge-base document.

Knowledge-base topic	**Expected relationship**
Bearing Temperature Monitoring	Observed 92 °C exceeds normal limit of 80 °C.
Abnormal Vibration	Observed 6.8 mm/s RMS requires recording and rotating-assembly/bearing checks.
Maintenance Evidence	Equipment identifier, observed value and inspection date should be recorded.
Lockout/Tagout	Maintenance on energized equipment requires isolation and verification before proceedi

7. Inspector Conclusion

The inspection identified abnormal bearing temperature, elevated vibration, and inadequate lubrication on P-101. The equipment should remain under controlled maintenance review until the bearing condition has been inspected and corrective actions have been completed in accordance with the applicable company procedures.

8. Approval / Review

Prepared by	Synthetic Inspection Team
Reviewed by	______________________________
Approved by	______________________________
Date	30 August 2026

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